

- 5 (a) (i) force = mg M1
in the direction of the field A1
- (ii) no force B1
- (b) (i) force due to E-field downwards so force due to B-field upwards M1
into the plane of the paper A1
- (ii) force due to magnetic field = Bqv B1
force due to electric field = Eq
- forces are equal (and opposite) so $Bv = E$ or $Eq = Bqv$ so $E = Bv$ B1
- (c) Component of velocity at right angle to the field results in a magnetic force at right angle to both this velocity and the field B1
radius $r = mv/Bq$, as B decreases, radius r increases
The particle describes a helical path with increasing radius and increasing pitch. B1

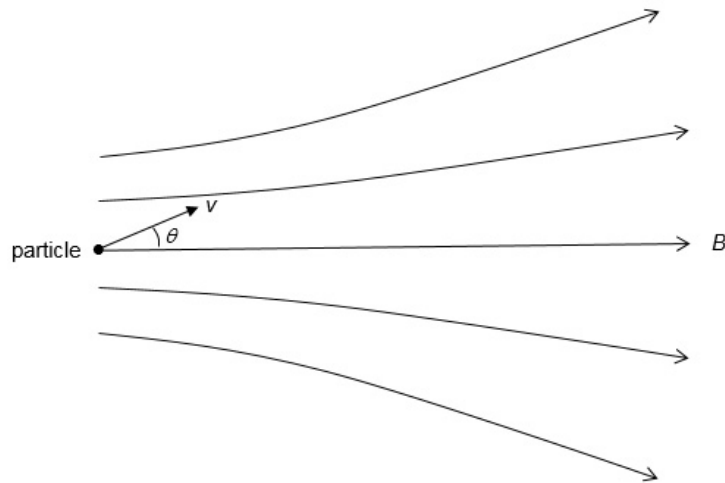


Diagram helical path with increasing radius B1
Correct Direction of path B1

6 (a) The binding energy of a nucleus is the energy released when a nucleus